

University of Nottingham

Mathematical Modeling Competition 2026

Competition Rules & Guidelines

1. Overview

The University of Nottingham Mathematical Modeling Competition 2026 invites undergraduate students from all three campuses (UNNC, UoN UK, and UNM) to tackle a challenging, real-world biomedical modeling problem. The use of AI tools is not only allowed but actively encouraged. However, success depends on your ability to think critically, reason biologically, and evaluate models—skills that AI cannot replicate on its own.

2. Eligibility

- Currently enrolled undergraduate students at UNNC, UoN UK, or UNM.
- Open to all interested students, regardless of department or prior coursework.
- Both individual participation and teams of up to 5 members are welcome. Teams are entirely self-organized. All members of a team must be from the same campus.

3. Registration

1. Complete the online registration form:
https://forms.office.com/Pages/ResponsePage.aspx?id=yMXEBIzbsUGIK1u3IIQF6GvdEX_qD6JGgOdkfZmcZnIUQzQxU0oxRzhBWIJRViY4RzgwMExZR1ZRQi4u
2. One form per team. One member submits on behalf of all.
3. Confirmation of registration will be sent via email within 48 hours.
4. Registered participants will receive the problem statement on 18 April 2026. Submissions are due by 17 May 2026.

4. Competition Timeline

Date	Activity
25 March 2026	Registration opens; team formation
18 April 2026	Problem statement released to all registered participants
18 April – 17 May	Student work period; Moodle forum for Q&A
17 May 2026, 23:59 GMT+8	Submission deadline
May – June 2026	Cross-campus evaluation by judging panel
Late June 2026	Results announced; certificates issued

5. Problem Statement

The problem statement will be released on 18 April 2026 to all registered participants. It focuses on antimicrobial resistance modeling in hospital settings—a complex, real-world

problem that requires both mathematical skill and biological understanding. Curated literature and software recommendations will be provided alongside the problem statement.

6. AI Use Policy

The use of AI tools is not only allowed but actively encouraged. You may use ChatGPT, DeepSeek, Claude, GitHub Copilot, or any other AI tool at any stage of your work. We want you to explore how AI can support mathematical modeling. However:

1. **AI Use Statement (mandatory):** You must submit a 1-page AI Use Statement documenting which AI tools you used, what tasks you delegated to AI, how you validated or modified AI outputs, and what you determined AI could not do well. A template is provided.
2. **Transparency is required:** Failure to disclose AI use, or submitting AI-generated content without acknowledgment, constitutes academic dishonesty and will result in disqualification.
3. **AI is a tool, not a substitute:** The evaluation rubric heavily weights skills AI cannot replicate (biological reasoning, model critique, creative synthesis). Teams that blindly copy AI outputs without demonstrating deep understanding will score poorly.

7. Submission Requirements

All submissions are made through the competition Moodle page. Required items:

- **Written Report (PDF, maximum 20 pages):** Must include mathematical model formulation, analysis, numerical simulations, validation, and critical evaluation.
- **Video Presentation (MP4, maximum 10 minutes):** Explaining your key findings and modeling decisions. This is a required component.
- **Simulation Code:** Upload all code files (Python, MATLAB, R, etc.) directly to Moodle.
- **AI Use Statement (PDF, 1 page, not counted in page limit):** Using the provided template.

8. Evaluation Criteria

Submissions will be evaluated by a judging panel using the following rubric:

Criterion	Weight	AI-Resistant?
Mathematical Framework	25%	Moderate
Biological Realism & Justification	25%	High
Analytical Depth	20%	Moderate
Critical Thinking & Model Evaluation	20%	High
Communication & Transparency	10%	Low

9. Prizes & Recognition

- All participants receive a certificate of participation.
- Winning teams receive a winners certificate.
- Winning teams will be announced on the competition website.

10. Academic Integrity

- Teams must work independently. Collaboration between teams is not permitted.
- All external sources (literature, data, code) must be properly cited.
- AI use must be fully disclosed in the AI Use Statement.
- Suspected academic dishonesty will be investigated and may result in disqualification.

11. Contact & Support

Moodle Forum: Use the competition Moodle forum for all technical and logistical questions.

Competition Organiser: Elio Espejo, School of Mathematical Sciences, UNNC

Email: z2019136@nottingham.edu.cn